

TRACE: Token Reconstruction and Availability-Aware Confidence-Weighted Ensemble for Incomplete Multimodal Learning

Shaowen Wan^a, Yanjun Lv^b, Lu Zhang^c, Dajiang Zhu^b, Bharat Biswal^a, Tianming Liu^d, Xiaobo Li^a and Lin Zhao^{a,*}

^aDepartment of Biomedical Engineering, New Jersey Institute of Technology, Newark, 07102, NJ, USA

^bDepartment of Computer Science and Engineering, The University of Texas at Arlington, Arlington, 76019, TX, USA

^cDepartment of Computer Science, Indiana University Indianapolis, Indianapolis, 46202, IN, USA

^dSchool of Computing, University of Georgia, Athens, 30602, GA, USA

ARTICLE INFO

Keywords:

Incomplete multimodal learning
Missing modality
Mixture-of-experts
Prediction ensemble
Neurobiological prediction

ABSTRACT

Multimodal neurobiological prediction can combine complementary information from imaging, genetics, clinical measurements, biospecimens, cognition, and behavior, but real cohorts contain heterogeneous subject-specific missingness. Existing methods commonly aggregate only the observed subset or reconstruct unavailable modalities, while paying less attention to how generated information should be distinguished and weighted during the final decision. We propose TRACE, a framework based on *Token Reconstruction and an Availability-aware Confidence-weighted Ensemble*. TRACE reconstructs each missing modality from the same subject's observed evidence, marks latent tokens as observed or generated, and estimates how much each candidate prediction should be trusted. A retained Transformer with sparse top- k experts models cross-modal interactions, while joint, unimodal, and pairwise branches are combined using availability, input reliability, shift-invariant predictive confidence, and a learned branch prior. Their reported probabilities and weights exactly reproduce the neural prediction. The final TRACE-CGHC predictor uses a frozen hierarchical consensus to reduce seed and model-family variance without replacing the sparse-MoE neural path. On three-class ADNI and ABCD tasks, it achieves the highest point estimate among six baselines for all six common metrics, including 67.30% and 62.01% accuracy, respectively. A separate five-seed, five-fold ADNI component study shows that latent completion and provenance marking improve macro-F1 within the conditional-generative neural component, while the dense-backbone control remains stronger on this development boundary. These results support provenance-aware reconstruction and trust-controlled decision fusion while retaining an explicit boundary between retrospective main comparisons and development-only ablations.

1. Introduction

Neurobiological phenotypes are characterized through heterogeneous sources that describe different levels of the same subject, including brain structure and function, genetic susceptibility, cognitive and clinical measurements, and behavioral or environmental factors [4, 6, 16]. These sources are not interchangeable: each may provide diagnostic information that is absent from the others, while their agreement can strengthen a prediction [31, 17, 20]. Multimodal learning therefore offers a natural basis for neurodevelopmental outcome prediction and neurodegenerative disease stratification. Its practical value, however, depends on whether a model can use the evidence that is actually available for each subject rather than assuming a fixed, complete input set.

This assumption is frequently violated in research cohorts and clinical data. Modalities can be absent because of acquisition cost, protocol differences, contraindications, participant dropout, or failed quality control, producing many subject-specific availability patterns [24, 29, 3, 10, 30]. Restricting analysis to complete cases discards otherwise informative subjects, whereas replacing every missing modality with a population-level constant removes the dependence

between the unavailable source and the subject's observed evidence. Robust incomplete-modality learning must consequently address two coupled questions: how to infer a task-relevant representation for an unavailable modality, and how strongly that inferred representation should influence the final prediction. Existing aggregation and alignment methods avoid explicit completion but may leave recoverable cross-modal information unused [21, 8]. Generative approaches estimate missing observations or features, but many are specialized to commensurate imaging modalities or optimized primarily for reconstruction fidelity [23, 19, 7, 32]. Flex-MoE supports arbitrary modality combinations through a missing-modality bank and sparse routing [28], yet its bank is indexed by the availability pattern and is therefore shared by subjects with the same observed subset. Less attention has been paid to jointly making completion subject-conditioned and making the downstream use of generated evidence reliability-aware.

Missingness also changes what constitutes a useful explanation. A sparse router indicates where computation is allocated, but its gate value is not, by itself, the contribution of a modality to the prediction. Likewise, post-hoc attribution can audit a trained model, yet it need not reproduce the decision rule used during inference [9, 18, 13, 15]. For an incomplete subject, an explanation must additionally distinguish an unavailable branch from an active but uncertain one,

*Corresponding author.

✉ lin.zhao.1@njit.edu (L. Zhao)

ORCID(s):

and it should not assign equal trust to directly observed and generated inputs. This motivates a decision mechanism in which availability, source reliability, predictive confidence, and branch contribution are represented explicitly and used in the prediction itself.

We propose TRACE, a unified framework for incomplete multimodal learning. Its name abbreviates *Token Reconstruction and an Availability-aware Confidence-weighted Ensemble*. The framework is organized around one information flow: *reconstruct, mark provenance, estimate trust, and make a decision*. TRACE maps heterogeneous modalities to a shared token space and reconstructs each unavailable target from a randomly sampled non-empty subset of the same subject's observed tokens. Source embeddings preserve whether a representation was observed or generated. A Transformer with sparse top- k experts serves as the cross-modal encoding backbone rather than as a separate conceptual module. TRACE then forms joint, unimodal, and pairwise diagnostic views and combines their probability outputs according to branch availability, input reliability, normalized-entropy confidence, and a low-capacity prior. The reported branch predictions and weights are exactly the quantities used to construct the neural prediction, providing a direct decision decomposition rather than a post-hoc surrogate.

We evaluate TRACE on two three-class cohorts: the Adolescent Brain Cognitive Development (ABCD) Study and the Alzheimer's Disease Neuroimaging Initiative (ADNI) [2, 22]. Both use four heterogeneous modality groups and retain naturally incomplete subjects. We additionally construct a fixed approximately 25% random-missing ABCD condition to isolate behavior under controlled partial-modality absence. The same six metrics are used for both datasets. This design motivates the following contributions:

- We formulate incomplete multimodal prediction as a coupled reconstruction-and-decision problem and introduce a unified framework that preserves the distinction between observed and generated evidence throughout the prediction path.
- We develop subject-conditioned latent reconstruction trained from stochastic observed subsets, with pooled semantic and normalized token-level supervision so that training contexts reflect arbitrary partial availability.
- We introduce a missingness-aware fusion mechanism that combines reliability and evidence over joint, unimodal, and pairwise branches. Its active branches, predictions, and normalized weights directly compose the final probability and can therefore be inspected without fitting an auxiliary explainer.
- We evaluate the framework with a common six-metric protocol on ADNI and ABCD, paired cluster-aware inference, and a matched five-seed, five-fold ADNI component study with a shared subject-level evaluation boundary.

2. Related Work

2.1. Representation Learning with Missing Modalities

Incomplete multimodal learning aims to exploit complementary sources without requiring every modality for every sample [24, 29]. Reconstruction-free methods operate directly on the observed subset. HeMIS aggregates available latent features through permutation-invariant statistics [3]; SMIL applies Bayesian meta-learning to severe missingness [10]; and M3Care models relations between patients and modalities in incomplete healthcare records [30]. Other methods constrain incomplete and complete views to occupy compatible representation spaces. ShaSpec separates shared and modality-specific factors and regularizes their distributions [21], whereas MissModal combines geometric, distributional, and semantic constraints across availability conditions [8]. These approaches avoid the cost and possible artifacts of synthesis, but their aligned or pooled representation does not explicitly state what information is supplied for a particular missing modality.

Generative methods instead recover absent observations or latent features. Multimodal variational autoencoders combine modality-specific posteriors to infer a shared latent variable [23], and generalized multimodal evidence lower bounds extend training to multiple observed subsets [19]. In incomplete neuroimaging, AGDiC couples adaptive graph learning, normalizing flows, and masked cross-attention to recover latent modality features under arbitrary missingness [7]. ACADiff uses clinically conditioned latent diffusion to synthesize missing MRI and PET volumes from the available images and metadata [32]. Such methods are well suited to modalities with meaningful cross-modal generation targets, but observation-level reconstruction can be unnecessarily expensive when the downstream task uses heterogeneous vector-valued sources. TRACE therefore reconstructs target-modality tokens rather than raw observations and supervises the generator on held-out representations from complete training subjects, focusing completion on a shared space used by the classifier.

Between these two families, pattern-conditioned completion offers a lighter alternative. Flex-MoE learns a missing-modality bank indexed by the observed-modality combination before applying sparse expert fusion [28]. This design supports arbitrary combinations efficiently, but every subject with the same availability pattern initially receives the same bank representation. TRACE instead conditions each target-modality reconstruction on the current subject's observed tokens. The generated tokens are also source-tagged, allowing downstream reliability estimation to distinguish them from direct observations.

2.2. Sparse Experts and Missingness-Aware Fusion

Conventional fusion applies essentially the same transformation to every sample even when the available modalities and their relevance differ. Sparse Mixture-of-Experts (MoE) introduces conditional computation by routing each

token to a small subset of expert networks. V-MoE demonstrated this principle at scale in vision Transformers [14], while MoE++ augments learned feed-forward experts with zero, copy, and constant experts to allocate both computation and transformations heterogeneously [5]. For incomplete multimodal learning, Flex-MoE combines a generalized router trained with complete samples and specialized routing for observed-modality combinations [28]. These methods establish sparse routing as a useful mechanism for representation specialization under heterogeneous inputs.

A related line of work structures experts around the type of cross-modal information being modeled. MMoE separates experts for unique, redundant, and synergistic information [27]. I²MoE adds weakly supervised interaction objectives and sample-specific reweighting to quantify these interaction types locally and globally [25]. This interaction-aware view is more informative than undifferentiated concatenation, but it addresses a different question from evidence reliability. Router or interaction weights determine how a representation is computed; they do not necessarily measure how much the resulting prediction should be trusted, especially when a modality is generated or absent. TRACE therefore uses sparse experts for feature transformation and a separate reliability–evidence mechanism for combining diagnostic branches.

2.3. Interpretability and Decision Decomposition

Clinical and neurobiological models are often audited with post-hoc feature attribution. SHAP, integrated gradients, and LIME estimate the sensitivity or local contribution of inputs under different reference, perturbation, or surrogate constructions [9, 18, 13]. Multimodal variants shift the unit of analysis from individual features to information sources: MM-SHAP quantifies modality contributions and can expose unimodal collapse [11], while Layer-Wise Modality Decomposition traces modality-specific contributions through a pretrained fusion network [12]. These methods are valuable for auditing arbitrary architectures, but the resulting attribution is external to the original prediction rule and may change with the explanation protocol.

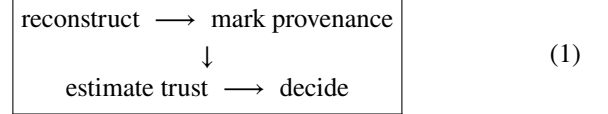
Interaction-based interpretation instead asks whether modalities contribute unique, redundant, or synergistic information. MMoE and I²MoE encode these categories in specialized experts [27, 25], and LSMI estimates sample-level interactions using pointwise information-theoretic quantities [26]. Such analyses characterize relationships among modalities, but they do not automatically decompose the probability produced by a classifier. This distinction becomes important under missingness: a pairwise interaction may be unavailable, and a branch supported by generated features may warrant less trust than the same branch supported by observations. TRACE addresses this decision-level requirement by explicitly predicting with joint, unimodal, and pairwise branches, masking unusable branches, and normalizing their contributions using modality reliability and predictive evidence. The resulting decomposition is faithful to the fusion rule in the limited but concrete sense

that the reported branch probabilities and weights exactly reconstruct the model’s final probability; it is not presented as a causal explanation of the underlying disease process.

3. Methodology

3.1. Problem Formulation and Design Principle

TRACE is organized around one principle: a missing modality should be reconstructed from the current subject, but the resulting representation must not be treated as if it had been observed. This leads to one connected information flow,



The stages answer three practical questions. First, how can an unavailable modality obtain a subject-specific representation? Second, how can the encoder retain whether that representation was measured or generated? Third, how much should a candidate prediction supported by that representation influence the final decision? Sparse MoE is retained as the interaction backbone between provenance marking and trust estimation; it is not introduced as a separate fifth stage.

Let $\mathcal{M} = \{1, \dots, M\}$ denote the modality set. For subject n , $\mathbf{x}_n^m$ is the input of modality m , $o_n^m \in \{0, 1\}$ indicates whether the entire modality is observed, and $y_n \in \{1, \dots, C\}$ is the class label. A modality-specific encoder maps each heterogeneous input to P tokens in a shared D -dimensional space:

$$\mathbf{Z}_n^m = E_m(\mathbf{x}_n^m), \quad \mathbf{Z}_n^m \in \mathbb{R}^{P \times D}. \quad (2)$$

The whole-modality mask is retained separately from missing entries inside an otherwise available table. Figure 1 summarizes how the same mask is used for reconstruction, provenance, and decision weighting.

3.2. Subject-Conditioned Completion and Provenance

A population-level missing token cannot adapt to the evidence of an individual subject. TRACE instead assigns each target modality m a conditional generator G_m . Let $\mathcal{O}_n = \{j : o_n^j = 1\}$ be the physically observed set. For a training subject with at least two observed modalities, we sample a non-empty proper context $\emptyset \neq S_n \subsetneq \mathcal{O}_n$ and use the remaining observed modalities $\mathcal{T}_n = \mathcal{O}_n \setminus S_n$ as reconstruction targets. The context for target $m \in \mathcal{T}_n$ is

$$\mathbf{C}_n^m = \text{Concat}(\{\mathbf{Z}_n^j : j \in S_n\}). \quad (3)$$

$$\hat{\mathbf{Z}}_n^m = G_m(\mathbf{Q}_m, \mathbf{C}_n^m), \quad \mathbf{Q}_m \in \mathbb{R}^{P \times D}, \quad (4)$$

where learned target queries pass through cross-attention and feed-forward refinement. A lightweight sigmoid output

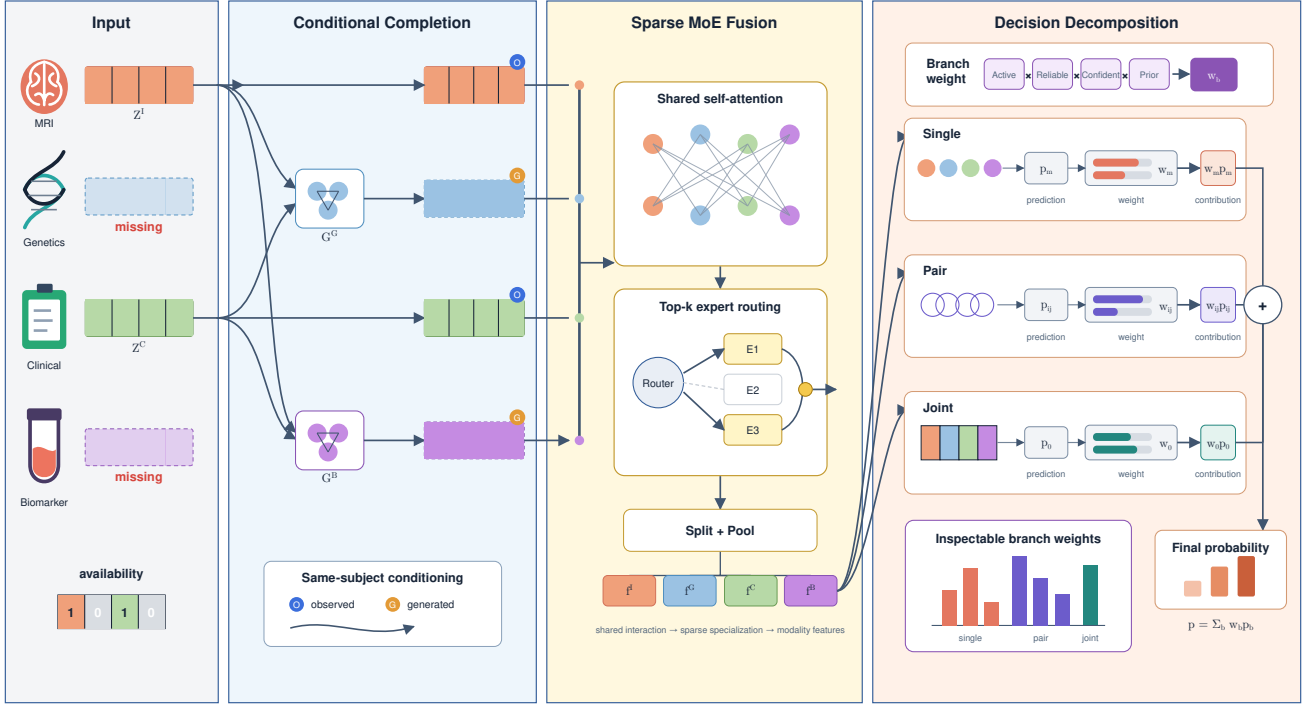


Figure 1: Overview of TRACE. Missing tokens are reconstructed from the same subject's observed modalities and explicitly marked as generated. A Transformer with sparse top- k experts produces task-aligned modality features. Unimodal, pairwise, and joint branches are then weighted by availability, input reliability, predictive confidence, and a branch prior. The displayed branch probabilities and weights exactly compose the neural prediction.

gate modulates the generated channels; it is an implementation detail rather than an additional conceptual module. At inference, $S_n = \mathcal{O}_n$, so every missing target is generated from all available evidence. Stochastic context sampling exposes the generator to both rich and sparse contexts instead of training only on leave-one-modality-out cases.

The reconstruction objective supervises both subject-level semantics and the full token sequence. Let $\bar{\mathbf{Z}}$ denote token averaging, R_m a target-specific projection, and $v(\mathbf{z}) = \mathbf{z}/(\|\mathbf{z}\|_2 + \epsilon)$ token normalization. We use

$$\begin{aligned} \mathcal{L}_{\text{rec}} &= \frac{1}{|\mathcal{R}|} \sum_{(n,m) \in \mathcal{R}} \left(\ell_{\text{pool}}^{n,m} + \lambda_{\text{tok}} \ell_{\text{tok}}^{n,m} \right), \\ \ell_{\text{pool}}^{n,m} &= 1 - \cos \left(R_m(\bar{\mathbf{Z}}_n^m), \text{sg} \left[R_m(\bar{\mathbf{Z}}_n^m) \right] \right), \\ \ell_{\text{tok}}^{n,m} &= \frac{1}{P} \sum_{p=1}^P \text{SmoothL1} \left(v(\hat{\mathbf{z}}_{n,p}^m), \text{sg} \left[v(\mathbf{z}_{n,p}^m) \right] \right), \end{aligned} \quad (5)$$

where $\mathcal{R}$ contains sampled eligible subject–target pairs and sg stops gradients through targets. The pooled term preserves subject-level content, while the token term prevents a multi-token reconstruction from being constrained only through its mean. Token indices are fixed latent positions created by the modality encoders and target queries; they are not interpreted as anatomical correspondence across modalities.

Completion alone does not communicate information status. Observed and generated tokens therefore receive different learned provenance embeddings:

$$\tilde{\mathbf{Z}}_n^m = \begin{cases} \mathbf{Z}_n^m + \mathbf{e}^{\text{obs}}, & o_n^m = 1, \\ \hat{\mathbf{Z}}_n^m + \mathbf{e}^{\text{gen}}, & o_n^m = 0. \end{cases} \quad (6)$$

The embeddings are shared across modalities. Completion therefore restores a common layout without erasing whether each token sequence was measured or inferred.

3.3. Missingness-Aware Multimodal Encoding

TRACE concatenates all provenance-marked tokens and applies global self-attention for cross-modal exchange. Its feed-forward transformation is a sparse MoE layer,

$$\begin{aligned} \mathcal{E}_k(\mathbf{h}) &= \text{TopK}(\mathbf{g}(\mathbf{h})), \\ \text{MoE}(\mathbf{h}) &= \sum_{e \in \mathcal{E}_k(\mathbf{h})} \pi_e(\mathbf{h}) F_e(\mathbf{h}), \\ \pi_e(\mathbf{h}) &= \frac{\exp(g_e(\mathbf{h}))}{\sum_{r \in \mathcal{E}_k(\mathbf{h})} \exp(g_r(\mathbf{h}))}. \end{aligned} \quad (7)$$

where F_e is expert e , $\mathbf{g}$ is a content-dependent router, and only the top- k experts are evaluated. Because provenance embeddings are part of each routed token, specialization can depend on both content and observed/generated status. A load-balancing loss discourages expert collapse. The resulting tokens $\mathbf{H}_n^m$ are summarized as $\mathbf{f}_n^m = \text{Pool}_m(\mathbf{H}_n^m)$ using a

stable mean with a small learned attentive residual. We treat sparse routing and pooling as backbone choices and evaluate the former against a dense control.

3.4. Reliability-Aware Decision Decomposition

Let $u_n^m = 1$ when modality m is observed or has been reconstructed from a non-empty context, and $u_n^m = 0$ otherwise. Thus a generated modality is usable but need not be trusted as strongly as an observation. The usability mask remains necessary when completion is disabled or no conditioning source exists. TRACE estimates modality reliability as

$$r_n^m = \begin{cases} 1, & o_n^m = 1, \\ \sigma(s_m^r(\mathbf{f}_n^m) + b_m^g), & m \text{ is generated,} \\ 0, & m \text{ is unusable,} \end{cases} \quad (8)$$

where b_m^g is initialized negatively. Observations provide the unit reference, whereas generated inputs receive subject- and modality-specific reliability.

The model constructs three prediction scales with distinct roles: unimodal branches capture source-specific evidence, pairwise branches capture low-order complementarity, and the joint branch captures global interaction. Let $\mathcal{B}$ contain one joint branch, M unimodal branches, and all $\binom{M}{2}$ pairwise branches. Their logits are

$$\begin{aligned} \mathcal{E}_n^0 &= F_0([\mathbf{f}_n^1; \dots; \mathbf{f}_n^M]), \\ \mathcal{E}_n^m &= F_m(\mathbf{f}_n^m), \\ \mathcal{E}_n^{ij} &= F_{ij}\left(\left[\mathbf{f}_n^i \odot \mathbf{f}_n^j; \mathbf{f}_n^i - \mathbf{f}_n^j\right]\right). \end{aligned} \quad (9)$$

The product and absolute-difference terms are computed only after the modality encoders and shared Transformer map every source to a task-aligned common space. They are learned interaction features, not biological coordinate correspondences. A branch is active only when its required inputs are usable. Its input quality is

$$\begin{aligned} q_n^0 &= \frac{\sum_m u_n^m r_n^m}{\sum_m u_n^m}, \\ q_n^m &= r_n^m, \quad q_n^{ij} = \sqrt{r_n^i r_n^j}. \end{aligned} \quad (10)$$

Input reliability and predictive confidence answer different questions: the former assesses the source representation, while the latter assesses the concentration of a branch prediction. For $\mathbf{p}_n^b = \text{softmax}(\mathcal{E}_n^b)$, TRACE uses detached normalized-entropy confidence,

$$\kappa_n^b = \max\left(1 - \frac{H(\text{sg}[\mathbf{p}_n^b])}{\log C}, \epsilon\right). \quad (11)$$

Unlike a confidence score based on unnormalized logit magnitude, Eq. (11) is invariant to adding the same constant to every class logit. It is a relative weighting signal, not a calibrated probability of correctness. The core trust rule is

therefore: trust a prediction when its input is reliable and its class distribution is concentrated. Combining activity A_n^b , quality, confidence, and a learned low-capacity prior β_b gives

$$\begin{aligned} s_n^b &= A_n^b(q_n^b + \epsilon)(\kappa_n^b + \epsilon) \exp(\beta_b), \\ w_n^b &= s_n^b / \sum_{r \in \mathcal{B}} s_n^r. \end{aligned} \quad (12)$$

The neural class probability is

$$\mathbf{p}_n = \sum_{b \in \mathcal{B}} w_n^b \text{softmax}(\mathcal{E}_n^b), \quad \mathcal{E}_n = \log(\mathbf{p}_n). \quad (13)$$

The stored branch probabilities and weights therefore reconstruct the neural output exactly. This is a faithful decomposition of the decision rule, not a causal explanation of disease mechanisms.

For exploratory visualization, branch mass can be allocated to modalities by assigning each unimodal weight to its source, splitting each pairwise weight equally, and distributing the joint weight uniformly:

$$d_n^m = w_n^m + \frac{1}{2} \sum_{j \neq m} w_n^{(m,j)} + \frac{w_n^0}{M}. \quad (14)$$

We refer to d_n^m as *modality decision allocation*. Unlike Eq. (13), this branch-to-modality split is a visualization convention, not a unique attribution. Token attention is likewise an auxiliary inspection signal rather than proof of feature importance.

3.5. Training Objective

TRACE is trained on an original view and an artificially reduced view that drops observed modalities while retaining at least one source. Rather than presenting every auxiliary loss as a separate module, we group the objective by role:

$$\begin{aligned} \mathcal{L}_{\text{task}} &= \mathcal{L}_{\text{cls}} + \lambda_b \mathcal{L}_{\text{branch}}, \\ \mathcal{L}_{\text{bal}} &= \mathcal{L}_{\text{gate}}^+ + \frac{1}{2} \mathcal{L}_{\text{gate}}^-, \\ \mathcal{L}_{\text{rob}} &= \lambda_d \mathcal{L}_{\text{drop}} + \lambda_s \mathcal{L}_{\text{sd}} + \lambda_m \mathcal{L}_{\text{MoFe}}, \\ \mathcal{L} &= \mathcal{L}_{\text{task}} + \lambda_r \mathcal{L}_{\text{rec}} + \lambda_g \mathcal{L}_{\text{bal}} + \mathcal{L}_{\text{rob}}. \end{aligned} \quad (15)$$

Here $\mathcal{L}_{\text{cls}}$ supervises the fused prediction, whereas $\mathcal{L}_{\text{branch}}$ averages cross-entropy over active branches and weights samples within each branch by detached branch quality. This prevents weak generated evidence from dominating auxiliary supervision.

Let $\mathcal{E}^+$ and $\mathcal{E}^-$ denote the original- and reduced-view logits. The reduced-view classification and self-distillation terms are

$$\begin{aligned} \mathcal{L}_{\text{drop}} &= \text{CE}(\mathcal{E}^-, y), \\ \mathcal{L}_{\text{sd}} &= T^2 \text{KL}(\text{sg}[\mathbf{p}_T^+] \parallel \mathbf{p}_T^-). \end{aligned} \quad (16)$$

The more-versus-fewer (MoFe) consistency term penalizes the original view only when its per-sample classification loss exceeds that of its strict reduced subset:

$$\mathcal{L}_{\text{MoFe}} = [\text{CE}(\mathcal{E}^+, y) - \text{CE}(\mathcal{E}^-, y)]_+. \quad (17)$$

Equation (15) retains MoE specialization while keeping the primary learning target explicit: reliable diagnosis under naturally incomplete and artificially reduced modality sets.

3.6. Frozen Hierarchical Consensus

The decision above is the output of one conditional-generative neural member. To reduce sensitivity to a single seed or fitted family, the reported predictor uses a fixed Conditional-Generative Heterogeneous Consensus (CGHC-v1). Coordinate-wise probability medians are computed across three seeds within each family and then across the family summaries; the result is normalized once onto the probability simplex. This hierarchy prevents a family with more checkpoints from receiving more influence merely because it contains more members.

For ABCD, the three families are conditional-generative Rank-MoE, CatBoost, and TabM-32; the neural family retains TRACE completion, provenance, sparse routing, and decision decomposition. ADNI uses four conditional-generation neural configurations, each with three seeds. The final consensus is deliberately a fixed median aggregation rather than another learned MoE gate. Hence sparse MoE remains part of the conditional-generative path, but the last cross-model operation is not itself an MoE. Within-member branch weights and cross-model aggregation have different meanings and are analyzed separately.

4. Experiments

4.1. Datasets

We evaluate TRACE on two three-class multimodal cohorts. Subjects are retained when at least one selected modality is available; complete-case filtering is not applied. **ABCD diagnostic burden.** The Adolescent Brain Cognitive Development (ABCD) Study is a longitudinal study of adolescent brain, cognitive, behavioral, and health development [2]. We predict three levels of lifetime K-SADS diagnostic-domain burden: no diagnosis, exactly one diagnosis domain, and two or more domains. This is an ordinal research phenotype, not a prospective clinical diagnosis. The family-disjoint cohort contains 11,867 participants, divided into 8,430 training, 1,692 validation, and 1,745 test participants. The four modality groups are imaging (structural MRI, resting-state fMRI, and diffusion MRI), genetic population structure, cognition and physical health, and non-target behavior and environment. All direct K-SADS target tables are excluded from the predictors.

ADNI. The Alzheimer's Disease Neuroimaging Initiative (ADNI) is a multicenter longitudinal study of biomarkers for cognitive decline and Alzheimer's disease [22]. We classify cognitively normal, mild cognitive impairment, and dementia. Four modalities are included: structural MRI-derived image features, genomic features, clinical assessments, and biospecimen measurements (IGCB). The cohort contains 2,116 participants, with 1,480, 318, and 318 assigned to the training, validation, and test splits, respectively.

Table 1

Frozen ABCD controlled-missingness cohort.

Split	Total	Complete	Missing	Missing (%)
Training	8430	6322	2108	25.01
Validation	1692	1269	423	25.00
Testing	1745	1309	436	24.99

4.2. Missingness Conditions

The main comparison preserves naturally incomplete inputs in both datasets. Whole-modality absence is recorded before model input construction, and a missing modality is never converted into an observed one by tabular imputation. To provide a controlled ABCD condition, we additionally freeze a participant-level mask that yields approximately 25% incomplete subjects in every split, including natural absence. Naturally complete participants are selected by label-independent hash ordering until the target rate is reached. Missing patterns are allocated across the 14 non-empty proper subsets of the four modalities, covering removal of one, two, or three modalities; all-four-missing cases are forbidden. The resulting counts are shown in Table 1. Every compared method within this condition uses exactly the same mask.

4.3. Preprocessing and Leakage Control

All feature selection, imputation, scaling, model selection, and threshold selection use training or validation data only. Test data are accessed after configurations are frozen. Whole-modality absence is recorded by a binary mask and remains distinct from missing entries inside an otherwise available modality.

For ABCD, each modality table is aligned to the immutable participant split. Numeric features exceeding the prespecified training-set missingness threshold or falling below the variance threshold are removed. Remaining within-modality entries are imputed with training-set medians and standardized using training-set statistics. Target-defining K-SADS tables are excluded, while non-K-SADS behavioral instruments remain predictors. Genetic-family members are assigned to only one split, although sites may occur across partitions; the experiment is family-disjoint but not site-held-out. In the controlled condition, modality masking occurs before preprocessing statistics are fitted.

For ADNI imaging, the latest FreeSurfer record per participant is retained according to its update timestamp. Structural features beginning with ST and ending in regional volume, thickness, or surface-related suffixes are selected, mean-imputed from training participants, and standardized with training statistics. Genotypes are filled using the training-set modal genotype and min-max transformed to $[-1, 1]$ using training-only extrema. Clinical fields associated with diagnosis timing are excluded, and missing Clinical and Biospecimen values are filled with training-set column means. An unavailable modality remains absent after within-modality preprocessing and is reconstructed only inside TRACE.

Table 2

Three-class ADNI performance under the common-six protocol. Values are percentages. Bold denotes the highest point estimate.

Method	Acc.	Bal. Acc.	Macro-F1	Wt.-F1	Macro-AUROC	Macro-AUPRC
Flex-MoE	62.89	61.21	60.50	61.30	79.63	66.07
I ² MoE	66.98	65.99	64.17	63.81	81.94	69.96
MoE++-corrected	63.52	63.42	62.85	63.27	80.96	67.57
AnyMod	64.15	63.09	62.43	63.33	81.97	70.83
AGDiC-inspired	58.81	55.64	56.74	58.23	76.56	62.20
ACADiff-inspired	56.60	54.20	52.04	53.25	71.74	56.78
TRACE-CGHC (Ours)	67.30	67.90	66.82	66.71	83.06	72.39

4.4. Baselines and Implementation

We compare TRACE with six incomplete-multimodal baselines. Flex-MoE uses an official-code-derived adaptation of its missing-modality bank and sparse routing [28]. I²MoE uses an official-code adapter for unique, redundant, and synergistic interaction experts [25]. MoE++-corrected is implemented through the I²MoE adapter with learned, zero, copy, and constant experts [5]. AnyMod is reimplemented from its modality-query and task-anchor formulation [1]. AGDiC-inspired and ACADiff-inspired are task-matched adaptations rather than direct reproductions of the original neuroimaging generation systems [7, 32]. These qualifiers are retained in the tables because differences in input representation and released code prevent the inspired methods from being interpreted as exact reproductions.

Each neural member uses $P = 16$ tokens per modality, hidden dimension $D = 128$, one fusion layer, four attention heads, 16 experts, and top-4 routing. Each target generator contains two cross-attention layers. Optimization uses AdamW with learning rate 10^{-4} , five warm-up epochs, gradient clipping at 5, modality-dropout probability 0.25, and normalized token-reconstruction weight 0.05. The principal auxiliary weights are $\lambda_g = 0.01$, $\lambda_r = 1.0$, $\lambda_b = 0.1$, $\lambda_d = 0.1$, $\lambda_s = 0.15$, and $\lambda_m = 0.1$. ABCD uses batch size 64; ADNI uses batch size 32. Dropout is 0.3. Models use seeds 0, 1, and 2, and all selection and fixed class-decision offsets are determined from validation data only. Experiments are executed on a single NVIDIA RTX 5070 Ti GPU.

4.5. Evaluation and Statistical Analysis

Both datasets use the same six metrics: accuracy, balanced accuracy, macro-F1, weighted-F1, macro-AUROC, and macro-AUPRC. The latter two are macro averages of one-versus-rest class scores. Each baseline is an equal three-seed probability ensemble. TRACE-CGHC uses the frozen hierarchy in Section 3.6. Paired inference is performed separately for accuracy and macro-F1 using 50,000 randomization draws and 20,000 one-sided 95% cluster-bootstrap draws. The cluster is the participant for ADNI and the genetic family for ABCD. Holm correction is applied across the six TRACE-to-baseline comparisons within each dataset and endpoint. Because the candidate and comparison boundary was retrospectively consolidated after model development, these tests are fixed-candidate paired evidence rather than a pristine prospective confirmatory experiment.

4.6. Main Results

TRACE-CGHC has the highest point estimate for every metric on both datasets. On ADNI (Table 2), the margins over the strongest baseline for the corresponding metric are 0.32 points in accuracy, 1.91 in balanced accuracy, 2.65 in macro-F1, 2.90 in weighted-F1, 1.09 in macro-AUROC, and 1.56 in macro-AUPRC. On ABCD (Table 3), the corresponding margins are 2.18, 0.62, 0.80, 1.69, 0.99, and 1.30 points. The gains are therefore consistent in direction across thresholded and ranking metrics, while their magnitudes differ by endpoint.

Multiplicity-corrected paired evidence is more selective than the point-estimate ranking. For ABCD accuracy, TRACE-CGHC is significantly better than each of the six baselines after Holm correction; adjusted p -values range from 1.2×10^{-4} to 9.0×10^{-3} , and every one-sided 95% cluster-bootstrap lower bound is positive. Against Flex-MoE specifically, the accuracy gain is 4.24 points (Holm $p = 1.8 \times 10^{-4}$, lower bound 2.67 points). ABCD macro-F1 is 1.63 points higher than Flex-MoE, but this difference is not significant after correction (Holm $p = 0.253$). On ADNI, the macro-F1 gain over Flex-MoE is 6.32 points (Holm $p = 0.031$, lower bound 2.12 points), whereas the 4.40-point accuracy gain is not Holm-significant ($p = 0.209$). We therefore distinguish the all-metric point-estimate result from endpoint-specific paired support.

4.7. ADNI Component Ablation Study

Table 4 reports the ADNI component study on 1,480 subjects using five subject-level out-of-fold splits and five seeds. Each control changes one declared factor while preserving the data boundary, optimization budget, and prediction aggregation. These values are development-only evidence for the conditional-generative neural component and must not be confused with the final heterogeneous TRACE-CGHC result in Table 2.

Removing latent completion lowers macro-F1 by 1.34 points and weighted-F1 by 1.08 points, although its ranking metrics are higher. Removing provenance marking lowers all six metrics, including macro-F1 by 0.46 points, while uniform branch weights lower macro-F1 by 0.22 points but improve macro-AUPRC by 1.06 points. The dense control is strongest on the four hard-decision metrics and improves macro-F1 by 2.19 points over the sparse-MoE configuration.

Table 3

Three-class family-disjoint ABCD diagnostic-burden performance under the common-six protocol. Values are percentages. Bold denotes the highest point estimate.

Method	Acc.	Bal. Acc.	Macro-F1	Wt.-F1	Macro-AUROC	Macro-AUPRC
Flex-MoE	57.77	54.26	54.53	58.78	74.69	57.69
I ² MoE	57.54	54.88	55.36	59.13	74.26	56.89
MoE++-corrected	59.83	54.56	54.13	59.15	74.32	57.52
AnyMod	59.60	54.19	54.78	59.51	73.43	56.64
AGDiC-inspired	56.79	53.55	53.73	57.76	71.81	54.94
ACADiff-inspired	57.25	54.19	54.67	58.61	73.17	56.53
TRACE-CGHC (Ours)	62.01	55.50	56.16	61.20	75.68	58.99

Table 4

ADNI matched component ablation on the five-fold development-OOF boundary. Values are percentages from the frozen five-seed campaign. Bold denotes the best value in each column.

Configuration	Acc.	Bal. Acc.	Macro-F1	Wt.-F1	Macro-AUROC	Macro-AUPRC
Full neural configuration	61.82	61.93	60.69	60.25	78.18	63.45
Without latent completion	61.62	61.69	59.36	59.17	79.58	66.25
Dense backbone	63.65	63.25	62.88	62.68	78.89	64.61
Without provenance marking	61.28	61.40	60.23	59.79	77.80	63.18
Uniform branch weights	60.74	61.14	60.48	59.89	78.33	64.51

Thus, this ADNI boundary supports completion and provenance marking for hard decisions, but does not establish a predictive advantage for sparse routing or reliability-aware weighting on every endpoint. Sparse MoE is retained as the interaction backbone of TRACE, while the dense row is reported transparently as a backbone-sensitivity result. No significance claim is made from this reused development boundary.

4.8. Decision-Allocation Analysis

Table 5 summarizes modality-level decision allocation from a validation-only replay of the frozen conditional-generative neural checkpoints. The rows are participant-level averages; the four entries sum to one. On ADNI, genomic allocation increases from 21.93% in cognitively normal participants to 25.91% in dementia, whereas image, clinical, and biospecimen allocations decrease. On ABCD, genetic and cognition/health allocation increase modestly with diagnostic burden. Cluster-bootstrap contrasts between the highest and lowest strata are Holm-significant for these reported changes except the ABCD imaging contrast. These associations describe how the fitted model distributes mixture mass; they are not etiologic or causal effects.

The exact probability-level decomposition is verified by replaying every retained neural checkpoint: the maximum absolute difference between the stored neural probability and the sum in Eq. (13) is zero. This verifies computational faithfulness of the branch mixture. It does not validate the heuristic allocation from branches to modalities in Eq. (14), which should be interpreted descriptively.

5. Discussion

The two cohorts expose complementary aspects of incomplete multimodal learning. ADNI combines imaging, genomic, clinical, and biospecimen evidence for cognitive-status classification; ABCD combines imaging, genetic, cognition/health, and behavior/environment evidence for diagnostic-burden classification. TRACE-CGHC is above every listed baseline on all six point estimates in both datasets. The cluster-aware analysis is intentionally more conservative: it supports ABCD accuracy against every baseline and ADNI macro-F1 against Flex-MoE, but it does not turn every positive point difference into a superiority claim.

TRACE combines conditional reconstruction, sparse experts, and weighted prediction branches in one missingness-aware pipeline. The contribution is the way information status is preserved across them: reconstruction is subject-conditioned, generated tokens are explicitly tagged and assigned learned reliability, and the final probability remains decomposable into active branch predictions. This is an architectural and decision-level contribution rather than a new theory of multimodal information decomposition. In particular, a large branch weight means that the branch had high relative quality and evidence under the learned fusion rule; it does not prove that the corresponding modality caused the outcome or represents a clinically valid biomarker.

The ADNI component study clarifies what the architecture contributes. Completion and provenance marking improve macro-F1 relative to their matched one-factor controls, but these gains are not uniform across ranking metrics. Uniform branch weighting produces a small macro-F1 reduction and a higher macro-AUPRC, while the dense backbone outperforms sparse MoE on this development

Table 5

Mean modality-level decision allocation (%) from the conditional-generative neural component.

Dataset	Stratum	Imaging	Genetic	Clinical or Cognition–health	Biospecimen or Behavior–environment
ADNI	Cognitively normal	23.22	21.93	28.51	26.35
ADNI	Mild cognitive impairment	23.42	22.28	28.41	25.89
ADNI	Dementia	22.05	25.91	27.26	24.78
ABCD	No diagnosis	23.80	26.24	23.03	26.92
ABCD	One diagnosis domain	23.76	26.64	23.16	26.44
ABCD	Two or more domains	23.63	26.95	23.31	26.11

boundary. The ablation therefore supports a focused, non-universal interpretation of individual design choices. The final heterogeneous consensus further reduces fitted-family variance, but its performance cannot be attributed solely to the neural architecture; this is why the component study is reported under a separate evaluation boundary.

Several limitations remain. First, normalized entropy is shift-invariant but not calibrated correctness; calibration error and reliability diagrams are still needed before clinical interpretation. Second, the ADNI component study reuses a development boundary, and the main paired comparison is retrospective fixed-candidate evidence rather than a newly preregistered confirmation. Third, ABCD is family-disjoint but not site-held-out, and external-cohort validation is absent. Natural missingness can encode acquisition and cohort processes, while the fixed-25% ABCD mask is label-independent but artificial; neither setting proves robustness to interventions that change which modalities are measured. Finally, AGDiC-inspired and ACADiff-inspired are task-matched adaptations rather than exact reproductions of their original imaging pipelines. These limitations motivate site-held-out evaluation, prospective freezing, calibration analysis, and replication on an independent cohort.

6. Conclusion

We presented TRACE as a coherent information-status pipeline: reconstruct missing latent tokens from subject-specific context, preserve their provenance, estimate the trust of candidate predictions, and aggregate those predictions transparently. Sparse MoE is retained as the interaction backbone, while a frozen hierarchical consensus stabilizes the reported predictor without masquerading as another MoE gate. TRACE–CGHC reaches 67.30% accuracy on ADNI and 62.01% on ABCD and has the highest point estimate across all six common metrics on both datasets. The ADNI component study supports latent completion and provenance marking for macro-F1, while the dense control’s stronger development result cautions against claiming that sparse routing is universally superior. The evidence supports the proposed information-status design while preserving a clear distinction between faithful prediction decomposition, descriptive modality allocation, retrospective paired comparisons, and development-only ablations.

References

- [1] Feng, Y., Gao, B., Deng, S., Qiu, A., Qin, J., 2024. Unified multimodal learning for any modality combinations in alzheimer’s disease diagnosis, in: Medical Image Computing and Computer Assisted Intervention – MICCAI 2024, Springer Nature Switzerland. pp. 487–497. doi:10.1007/978-3-031-72384-1_46.
- [2] Garavan, H., Bartsch, H., Conway, K., Decastro, A., Goldstein, R.Z., Heeringa, S., Jernigan, T., Potter, A., Thompson, W., Zahs, D., 2018. Recruiting the ABCD sample: Design considerations and procedures. *Developmental Cognitive Neuroscience* 32, 16–22. doi:10.1016/j.dcn.2018.04.004.
- [3] Havaei, M., Guizard, N., Larochelle, H., Jodoin, P., 2016. Hemis: Hetero-modal image segmentation. *Medical Image Computing and Computer-Assisted Intervention*.
- [4] Jack, C.R., Bennett, D.A., Blennow, K., Carrillo, M.C., Dunn, B., Haeberlein, S.B., Holtzman, D.M., Jagust, W., Jessen, F., Karlawish, J., et al., 2018. NIA-AA research framework: Toward a biological definition of alzheimer’s disease. *Alzheimer’s & Dementia*.
- [5] Jin, P., Zhu, B., Yuan, L., Yan, S., 2025. MoE++: Accelerating mixture-of-experts methods with zero-computation experts, in: International Conference on Learning Representations.
- [6] Lane, C.A., Hardy, J., Schott, J.M., 2018. Alzheimer’s disease. *European Journal of Neurology*.
- [7] Li, Y., Dong, Y., Wu, Y., Yan, H., Gao, L., 2025. Alzheimer’s disease recognition based on adaptive graph normalization flow for incomplete multimodal data fusion, in: Medical Image Computing and Computer Assisted Intervention – MICCAI 2025, Springer Nature Switzerland. pp. 64–73. doi:10.1007/978-3-032-04984-1_7.
- [8] Lin, R., Hu, H., 2023. MissModal: Increasing robustness to missing modality in multimodal sentiment analysis. *Transactions of the Association for Computational Linguistics* 11, 1686–1702. doi:10.1162/tac1_a_00628.
- [9] Lundberg, S.M., Lee, S., 2017. A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*.
- [10] Ma, M., Ren, J., Zhao, L., Tulyakov, S., Wu, C., Peng, X., 2021. Smil: Multimodal learning with severely missing modality. *Proceedings of the AAAI Conference on Artificial Intelligence*.
- [11] Parcalabescu, L., Frank, A., 2023. MM-SHAP: A performance-agnostic metric for measuring multimodal contributions in vision and language models and tasks, in: Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics, Association for Computational Linguistics. pp. 4032–4059. doi:10.18653/v1/2023.acl-long.223.
- [12] Park, J., Park, K., Kim, D., Park, J., Choi, J.W., 2025. Layer-wise modality decomposition for interpretable multimodal sensor fusion, in: *Advances in Neural Information Processing Systems*.
- [13] Ribeiro, M.T., Singh, S., Guestrin, C., 2016. “why should i trust you?": Explaining the predictions of any classifier. *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining*.
- [14] Riquelme, C., Puigcerver, J., Mustafa, B., Neumann, M., Jenatton, R., Pinto, A.S., Keysers, D., Houlsby, N., 2021. Scaling vision

- with sparse mixture of experts, in: *Advances in Neural Information Processing Systems*.
- [15] Rudin, C., 2019. Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nature Machine Intelligence*.
 - [16] Scheltens, P., De Strooper, B., Kivipelto, M., Holstege, H., Chéte-
lat, G., Teunissen, C.E., Cummings, J., van der Flier, W.M., 2021. Alzheimer's disease. *The Lancet*.
 - [17] Suk, H.I., Lee, S.W., Shen, D., Alzheimer's Disease Neuroimaging Initiative, 2014. Hierarchical feature representation and multimodal fusion with deep learning for ad/mci diagnosis. *NeuroImage*.
 - [18] Sundararajan, M., Taly, A., Yan, Q., 2017. Axiomatic attribution for deep networks. *International Conference on Machine Learning*.
 - [19] Sutter, T.M., Daunhawer, I., Vogt, J.E., 2021. Generalized multimodal elbo. *International Conference on Learning Representations*.
 - [20] Venugopalan, J., Tong, L., Hassanzadeh, H.R., Wang, M.D., 2021. Multimodal deep learning models for early detection of alzheimer's disease stage. *Scientific Reports*.
 - [21] Wang, H., Chen, Y., Ma, C., Avery, J., Hull, L., Carneiro, G., 2023. Multi-modal learning with missing modality via shared-specific feature modelling, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 15878–15887.
 - [22] Weiner, M.W., Veitch, D.P., Aisen, P.S., Beckett, L.A., Cairns, N.J., Green, R.C., Harvey, D., Jack, C.R., Jagust, W., Liu, E., et al., 2013. The alzheimer's disease neuroimaging initiative: A review of papers published since its inception. *Alzheimer's & Dementia*.
 - [23] Wu, M., Goodman, N.D., 2018. Multimodal generative models for scalable weakly-supervised learning. *Advances in Neural Information Processing Systems*.
 - [24] Wu, R., Wang, H., Chen, H.T., Carneiro, G., 2024. Deep multi-modal learning with missing modality: A survey. *arXiv preprint arXiv:2409.07825*.
 - [25] Xin, J., Yun, S., Peng, J., Choi, I., Ballard, J.L., Chen, T., Long, Q., 2025. I^2 MoE: Interpretable multimodal interaction-aware mixture-of-experts, in: *Proceedings of the 42nd International Conference on Machine Learning*, PMLR. pp. 68870–68888.
 - [26] Yang, Z., Wang, H., Hu, D., 2025. Efficient quantification of multimodal interaction at sample level, in: *Proceedings of the 42nd International Conference on Machine Learning*, PMLR. pp. 71302–71317.
 - [27] Yu, H., Qi, Z., Jang, L.K., Salakhutdinov, R., Morency, L.P., Liang, P.P., 2024. MMoE: Enhancing multimodal models with mixtures of multimodal interaction experts, in: *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, Association for Computational Linguistics. pp. 10006–10030. doi:10.18653/v1/2024.emnlp-main.558.
 - [28] Yun, S., Choi, I., Peng, J., Wu, Y., Bao, J., Zhang, Q., Xin, J., Long, Q., Chen, T., 2024. Flex-moe: Modeling arbitrary modality combination via the flexible mixture-of-experts. *Advances in Neural Information Processing Systems*.
 - [29] Zhan, Y., Yang, R., You, J., Huang, M., Liu, W., Liu, X., 2025. A systematic literature review on incomplete multimodal learning: techniques and challenges. *Systems Science & Control Engineering*.
 - [30] Zhang, C., Chu, X., Ma, L., Zhu, Y., Wang, Y., Wang, J., Zhao, J., 2022. M3care: Learning with missing modalities in multimodal healthcare data. *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*.
 - [31] Zhang, D., Wang, Y., Zhou, L., Yuan, H., Shen, D., Initiative, A.D.N., et al., 2011. Multimodal classification of alzheimer's disease and mild cognitive impairment. *NeuroImage*.
 - [32] Zhou, R., Zhou, H., Su, Y., Chen, B.Y., Zhang, Y., He, L., Alzheimer's Disease Neuroimaging Initiative, 2026. Adaptive clinical-aware latent diffusion for multimodal brain image generation and missing modality imputation. *arXiv preprint arXiv:2603.09931* doi:10.48550/arXiv.2603.09931.